

CPR is as easy as
C-A-B



Compressions

Push hard and fast
on the center of
victim's chest



Airway

Tilt the victim's head
back and lift the chin
to open the airway



Breathing

Give mouth-to-mouth
rescue breaths

American Heart
Association

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Cardiopulmonary Resuscitation

*Prepared by/
Dr/ Naglaa Nassar*

Cardiopulmonary Resuscitation

Introduction

Codes are called when patients have a cardiac and/or respiratory arrest or a life-threatening cardiac dysrhythmia that causes a loss of consciousness.

Code management refers to the initiation of a code and the lifesaving interventions performed when a patient arrests .

Rapid Response Teams (RRTs)

Have recently been implemented to address changes in a patient's clinical condition before a cardiac and/or respiratory arrest occurs. The intent of a RRT is preventing the cardiac and/or respiratory arrest, from ever occurring. The role of RRTs in preventing cardiopulmonary arrest, the roles of the personnel involved in a code, and equipment that must be readily available to support interventions of the RRT or during a code. Basic and advanced life support measures are presented, including medications commonly used during a code.

Definition of Cardiac Arrest

Cardiac arrest is defined as abrupt cessation of effective cardiac pumping activity, resulting in cessation of circulation. Unless definitive action is taken within 4 to 6 minutes, the patient will suffer irreversible brain injury.

Cardiopulmonary Resuscitation (CPR)

Resuscitation is a restoration of vital signs by mechanical, physiological, and pharmacological means.

Code is an informal term for emergency resuscitation. Code management refers to the initiation of a code and the lifesaving interventions performed when a patient arrests

Causes of Cardiopulmonary Arrest

- **Cardiac Causes**
- **Pulmonary Causes**
- **Electrolyte Imbalances Causes**
- **Procedural Causes**

Causes of Cardiopulmonary Arrest

Cardiac Causes

- Myocardial Infarction
- Heart Failure
- Dysrhythmia
- Coronary artery spasms
- Cardiac tamponade

Pulmonary Causes

- Respiratory failure secondary to respiratory depression
- Airway obstruction
- Acute Respiratory Distress Syndrome (ARDS)
- Impaired ventilation, such as pneumothorax

Cont, Causes of Cardiopulmonary Arrest

Electrolyte Imbalances Causes

- Hyperkalemia
- Hypomagnesemia
- Hypercalcemia/hypocalcemia

Procedural Causes

- Pulmonary artery catheterization
- Cardiac catheterization
- Surgery

Basic Life Support (BLS)

1. Check Responsiveness

- 1) The nurse first determines the patient's responsiveness before initiating CPR. Tap and shout, 'Are you all right?'
 1. The nurse should take no more than 10 seconds to check for breathing and pulse before initiating CPR.
 2. The nurse calls for help ("initiate a code") and initiates BLS measures. In the new American Heart Association (AHA) Guidelines for CPR, the new acronym to guide resuscitative efforts is C-A-B, rather than A-B-C as in previous years



Check Circulation, Airway and Breathing

The patient should be placed in a supine position on a firm, flat surface. A back or "cardiac" board is usually located on the back or side of the cart. It is placed under the patient as soon as possible to provide a hard, level surface for the performance of chest compressions

Circulation

- Compress chest at rate of at least 100/min
- Compress chest to a depth at least 2 inches
- Place backboard or use CPR setting on bed
- Watch ECG monitor to assist with ensuring rate
- Palpate pulses (radial, femoral, pedal) to determine effectiveness of compressions
- Measure blood pressure manually

- The arrival of a second rescuer allows for compressions to continue while the second rescuer can open the airway and be ready to deliver breaths via a bag-valve device when the initial rescuer has completed a 30 chest compression cycle. The chest must be allowed to fully recoil between compressions. This rhythm is maintained at the rate of at least 100 times/min.

Airway

Open patients airway using head tilt-chin lift maneuver (jaw thrust for cervically injured patients). If spontaneous respirations have not returned once a patent airway has been established, then the patient must be assisted with breathing. head tilt-chin lift maneuver



Breathing

- Use Ambu bag, set for 100% oxygen (oxygen source set at 15 L/min).
- Aim for compression to breath ratio 30:2.
- Deliver breaths at rate of 6-8/min once advanced airway is in position.
- Maintain seal around patient's mouth and nose.
- Observe for chest rise and fall.
- A second person assisting with CPR, auscultate all lung fields to confirm that the delivered breaths are reaching the lungs. Pulse oximetry is used to determine oxygenation.

Advanced Cardiac Life Support (ACLS)

Airway

Airway management involves reassessment of original techniques established in BLS. Endotracheal intubation provides definitive airway management.

Breathing

Breathing assessment determines whether the ventilatory efforts are causing the chest to rise. If the patient is intubated, ETT placement is confirmed by bilateral breath sounds and observation of chest movement with ventilation. A chest radiograph confirms placement after the code.

Circulation

- Circulation initially focuses on chest compressions, attachment of electrodes and leads to the monitor-defibrillator, rhythm identification, IV access, and medication administration.
- If ventricular fibrillation or pulseless ventricular tachycardia is identified, high energy unsynchronized shocks are delivered followed by two minutes of chest compressions with ventilation.
- A patent IV is necessary during an arrest for the administration of fluids, medications, or both.

Role of Resuscitation Team Members

When a cardiopulmonary arrest (“code”) is called, various members of the emergency response team are notified. Key personnel are notified to assist with code management

Director of the Code (Physician/Nurse Practitioner/ Advanced Cardiac Life Support (ACLS) Qualified Personnel)

Role of Resuscitation Team Members

Roles and Responsibilities of Code Team Members	
Team Member	Primary Role
Leader of the code (usually a physician)	Directs code
	Makes diagnoses and treatment decisions
Primary nurse	Provides information to code leader
	Measures vital signs
	Assists with procedures
	Administers medications
Second nurse	Coordinates use of the crash cart
	Prepares medications
	Assembles equipment (intubation, suction)
Nursing supervisor	Controls the crowd
	Contacts the attending physician
	Assists with medications and procedures
	Ensures that a bed is available in critical care unit (ICU)
	Assists with transfer of the patient to the ICU
Nurse or assistant	Records events on designated form
Anesthesiologist or nurse anesthetist	Intubates patient
	Manages airway and oxygenation
Respiratory therapist	Assists with ventilation and intubation
	Obtains blood sample for ABG analysis
	Sets up respiratory equipment/mechanical ventilator
Pharmacist or pharmacy technician	Assists with medication preparation

Equipment used in Cardiopulmonary Arrest

While the first person to recognize a code calls for help and begins life support measures, another team member immediately brings the crash cart and defibrillator to the patient's bedside. The carts must be inventoried daily to ensure their contents are complete and available in the event of a cardiopulmonary arrest.

Contents of a Crash Cart

MAIN ITEMS	SPECIFIC SUPPLIES
Back	
Cardiac board	
Side	
Portable suction machine, bag-valve device, and oxygen tubing	Suction device with canister and tubing, face mask, and oxygen tank container for disposing of needles, syringes, and other sharp items
Top	
Monitor-defibrillator with recorder, clipboard with code record and drug calculation reference sheets	ECG leads, electrodes, conductive gel or adhesive electrode pads; possible transcutaneous pacemaker or combination unit
Airway equipment drawer or box	Oral and nasal airways, ETTs, laryngoscope handle and curved and straight blades, lubricating jelly, 10 mL syringes, and tape
IV equipment drawer	IV catheters of various sizes, tape, syringes, needles, IV fluids (NS, Ringer's'lactate solution, and D5W); and IV tubing
Medication drawer or box	All IV push emergency medications in prefilled syringes if available, sterile water and NS for injection, and IV infusion emergency medications
Miscellaneous supply drawer	Sterile and non sterile gloves, suction catheters, nasogastric tubes, chest tubes, blood pressure cuff, blood collection tubes, sutures, pacemaker magnet, extra ECG
Procedure kits	recording paper, gauze pads, face masks
	Arterial blood gases, tracheotomy, , central line insertion kit, chest tube tray

Contents of a Crash Cart

✓ Defibrillator

✓ Breathing equipment/
air supplies

✓ Emergency drugs

✓ IV supplies
and tubing



Medications used to treat a patient in Cardiac arrest

Medications that are administered during a code depend on several factors:

cause of the arrest, patient's cardiac rhythm, physician's preference, patient's response.

Goals of Treatment

- * Reestablish and maintain optimal cardiac function
- * Correct hypoxemia and acidosis
- * Suppress dangerous cardiac ectopic activity
- * In addition, medications are used to achieve a balance between myocardial oxygen supply and demand
- * Maintain adequate blood pressure, and to relieve heart failure.

1. Epinephrine (Adrenalin)

Mechanism of action

- Because of its alpha-adrenergic and beta-adrenergic Effect. Epinephrine is a potent vasoconstrictor. Epinephrine increases systemic vascular resistance and arterial blood pressure, as well as heart rate, contractility, and automaticity of cardiac pacemaker cells.

Indication

- Indicated for the restoration of cardiac electrical activity in an arrest
- Epinephrine is used to treat VF or pulseless VT that is unresponsive to initial defibrillation, asystole, and PEA.
- During a code, epinephrine may be given IV

Dosage

- The IV dosage is 1.0 mg and is repeated every 3 to 5 minutes as needed
- Epinephrine may be administered by continuous infusion to increase the heart rate or blood pressure.
- (N.B) because epinephrine increases myocardial oxygen requirements, the nurse must monitor the patient closely for signs of myocardial ischemia.

Side effect

- Tachycardia
- Hypertension

2. Vasopressin a potent vasoconstrictor

Action

- ❖ Vasopressin is recommended as an alternative to Epinephrine administration in VF unresponsive to shock.

Dose

- ❖ A single dose of 40 units IV
- ❖ Administration of the drug is not recommended for conscious patients with coronary artery disease because severe angina can result from the vasoconstriction.

Side effect

Myocardial ischemia

3. Atropine (Anticholinergic agent)

Action

Atropine is used to increase the heart rate by decreasing the vagal tone

Indication

- ❖ It is indicated for patients with symptomatic bradycardia.
- ❖ For symptomatic bradycardia, atropine 0.5 mg IV is given and repeated every 3 to 5 minutes as needed (for a total of 3 mg) to maintain a heart rate greater than 60 beats per minute or until adequate tissue perfusion is achieved as indicated by blood pressure and level of consciousness.
- ❖ In asystole a single dose of 3 mg should be given .

Side Effect

- ❖ Atropine is used cautiously in acute coronary ischemia or myocardial infarction because the increased heart rate may worsen ischemia or increase infarction size.

4. Amiodarone (Cordarone) Antiarrhythmic class

Action

Decrease membrane excitability, prolongs action potential to terminate VT . dilating coronary arteries and increasing coronary blood supply (potent vasodilator)

Indication

- ❖ Treatment and prophylaxis of recurrent VF : and hemodynamically unstable VT; rapid atrial dysrhythmias
- ❖ Cardiac Arrest Dose: 300 mg IV push followed by a dose of 150 mg

Recurrent VF/VT: 150 mg IV over 10 minutes (15 mg per minute); may repeat 150 mg IV every 10 minutes as needed followed by 360 mg infusion for 6 hours

Side effect

- * Bradycardia
- * Hypotension

5. Sodium Bicarbonate

- ❖ A patient who has experienced an arrest quickly becomes acidotic.
- ❖ Acidosis results from two sources: no blood flow during the arrest, and low blood flow during CPR.
- ❖ Effective ventilation with supplemental oxygen and rapid restoration of tissue perfusion by CPR and spontaneous circulation are the best mechanisms to correct these causes of acidosis.
- ❖ The initial dosage of sodium bicarbonate is 1 mEq/kg by IV push

6. Dopamine (Intropin)

Indication

- symptomatic hypotension in the absence of hypovolemia.
- It is the second line medication for symptomatic bradycardia (after atropine).

Dosage

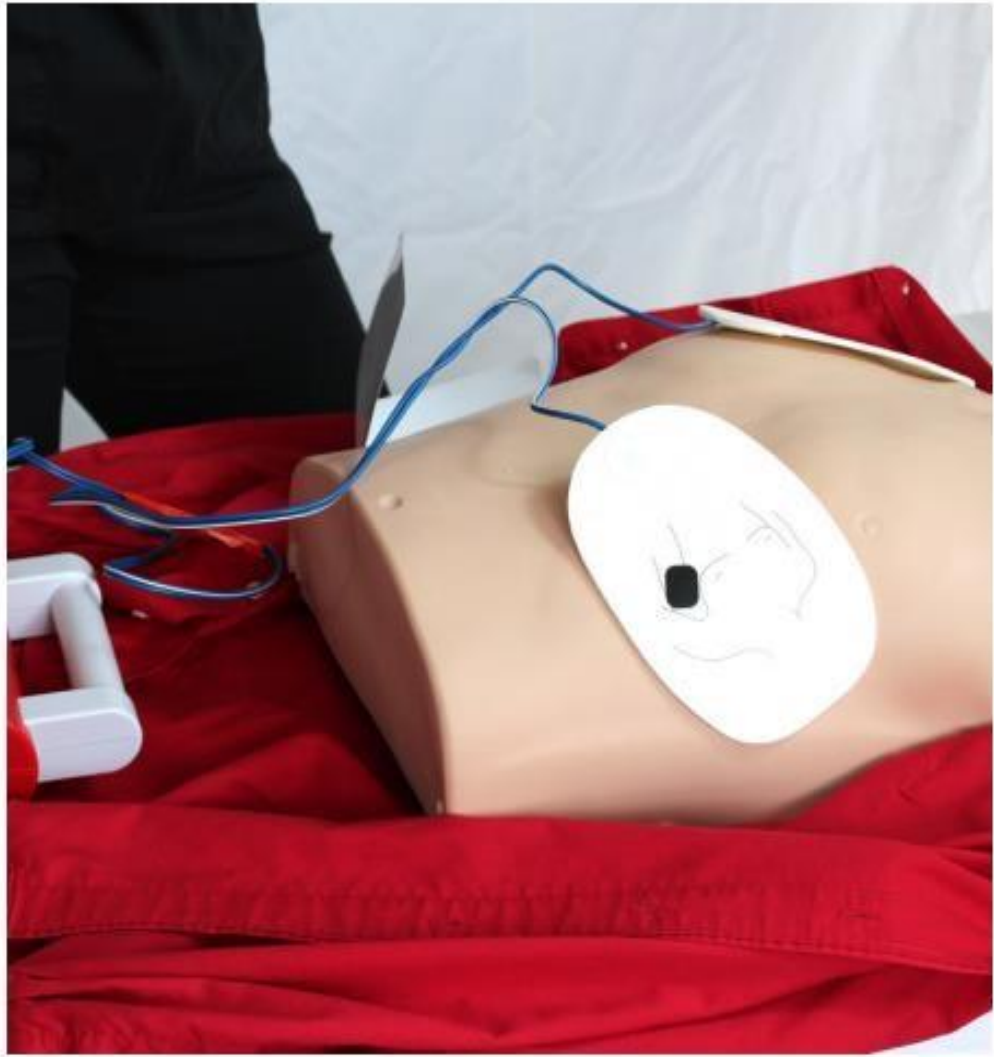
- Dopamine is administered by continuous IV infusion starting at 2 to 5 mcg/kg/min, and the dose is titrated upward according to patient response.
- A dilution of 400 to 800 mg of dopamine in 250 to 500-mL of D5W delivers 1600 to 3200 mcg/mL.

Defibrillation

Defibrillation: It is the delivery of an electrical current to the heart through the use of a defibrillator. The current can be delivered through the chest wall by use of external paddles or adhesive electrode pads connected to cables.

All patients in an ICU are connected to a cardiac monitor that provides useful information during a cardiopulmonary arrest. Once BLS procedures have been initiated, further intervention may be necessary.

The nurse assesses the ECG rhythms on the monitor throughout the resuscitation. If the patient is in VF or pulseless VT, preparations for defibrillation need to be started immediately to avoid death.



The delivery of an electrical impulse by an external defibrillator simultaneously **depolarizes most of the ventricular cells during VF and the reentry abnormalities of VT.**

Following this first shock, **five cycles of CPR should be performed. If the patient remains in a shockable rhythm then a second shock should be delivered.** After each subsequent shock, five more cycles of CPR should be performed before assessing whether the patient has remained in a shockable rhythm.

Automated External Defibrillation (AED)

The AED is an external defibrillator with rhythm analysis capabilities.

It is used to achieve early defibrillation.

Cardioversion

Cardioversion is the delivery of a shock that is synchronized with the patient's cardiac rhythm. The purpose of Cardioversion is to disrupt an ectopic pacemaker that is causing a dysrhythmia and to allow the sino atrial node to take control of the rhythm.

Cardioversion is similar to defibrillation, except the delivery of energy is synchronized to occur during ventricular depolarization.

Care of the patient after Resuscitation

Post resuscitation goals include

1. Optimizing cardiopulmonary function and tissue perfusion,
2. Transporting the patient to an appropriate critical care unit capable of providing post cardiac arrest care
3. Identifying and treating the precipitating causes of the arrest to help prevent another arrest.

Management of post cardiac arrest

- Advanced airway placement
- Maintenance of blood pressure and oxygenation
- Control of dysrhythmias
- Advanced neurological monitoring
- Use of therapeutic hypothermia.

1. The unconscious patient **will require an advanced airway or endotracheal intubation.**
2. **Continuous ECG monitoring** continues after Return Of Spontaneous Circulation (ROSC), during transport to the critical care unit
3. **IV access is obtained** if not already established and position and function of any IV catheter is verified
4. If the patient is hypotensive (systolic blood pressure less than 90 mm Hg) fluid boluses of 1 to 2 liters **normal saline or lactated Ringer solution are considered.**
5. **Vasoactive medications** including Dopamine (5 to 10 mcg/kg/min), norepinephrine (0.1 to 0.5 mcg kg/min) or epinephrine (1 to 10 mcg/min) maybe initiated.
6. **Serial neurological exams** are performed, including response to verbal commands or physical stimulation, pupillary response to light, presence of corneal reflex, spontaneous eye movement, gag, cough, and spontaneous breaths.

- ❖ Intermittent or continuous IV sedation, analgesia, or both can be used to relieve pain, discomfort, and anxiety.
- ❖ Serial head computed tomography (CT) scans and electroencephalography (EEG) monitoring may be performed if the patient is comatose and/or has seizures.
- ❖ Hyperglycemia may occur post cardiac arrest. The optimum blood glucose concentration and strategies are not known for patient post cardiac arrest. However, a target range of 144 to 180 mg/dL has been shown to decrease the risk of hyperglycemia after cardiac arrest.

❖ Emotional support is an important aspect of care after an arrest. Fear of death or of a recurrence of the arrest is common.

❖ Nurses should listen objectively and provide psychological support. In addition to the patient, many other people are affected when a code occurs



Thank You!